

Problem 12

Problem 12

Problem 1 Evaluate the following indefinite integrals:

(a) $\int \frac{13x^7}{\sqrt{3x^4 - 5}} dx$

Explanation. Set $v = 3x^4 - 5$. Then $dv = 12x^3 dx$ and $x^4 = \frac{1}{3}(v + 5)$. Substituting into the integral gives:

$$\begin{aligned} \int \frac{13x^7}{\sqrt{3x^4 - 5}} dx &= \frac{13}{12} \int \frac{(x^4)(12x^3)}{\sqrt{3x^4 - 5}} dx \\ &= \frac{13}{12} \int \frac{\frac{1}{3}(v + 5)}{\sqrt{v}} dv \\ &= \frac{13}{36} \int \left(v^{\frac{1}{2}} + 5v^{-\frac{1}{2}} \right) dv \\ &= \frac{13}{36} \left(\frac{2}{3} v^{\frac{3}{2}} + 10v^{\frac{1}{2}} \right) + C \\ &= \frac{13}{36} \left(\frac{2}{3} (3x^4 - 5)^{\frac{3}{2}} + 10(3x^4 - 5)^{\frac{1}{2}} \right) + C. \end{aligned}$$

(b) $\int \frac{u^3}{u^2 - 3} du$

Explanation. Set $w = u^2 - 3$. Then $dw = 2u du$ and $u^2 = w + 3$. This gives

$$\begin{aligned} \int \frac{u^3}{u^2 - 3} du &= \frac{1}{2} \int \frac{(u^2)(2u)}{u^2 - 3} du \\ &= \frac{1}{2} \int \frac{w + 3}{w} dw \\ &= \frac{1}{2} \int \left(1 + \frac{3}{w} \right) dw \\ &= \frac{1}{2} (w + 3 \ln |w|) + C \\ &= \frac{1}{2} (u^2 - 3 + 3 \ln |u^2 - 3|) + C \end{aligned}$$

(c) $\int e^{t^2 + \ln(t)} dt$

Explanation. First notice that $\int e^{t^2+\ln(t)} dt = \int e^{t^2} e^{\ln(t)} dt = \int te^{t^2} dt$.

Let $u = t^2$. Then $du = 2t dt$ and $\frac{1}{2} du = t dt$.

$$\begin{aligned} \int e^{t^2+\ln(t)} dt &= \int te^{t^2} dt \\ &= \frac{1}{2} \int e^u du \\ &= \frac{1}{2} e^u + C \\ &= \frac{1}{2} e^{t^2} + C. \end{aligned}$$